

A correlation and risk monitoring blueprint for a 3-7 strategy retail book

A solo operator running 3-7 equity strategies on Alpaca does not need DCC-GARCH or copulas to stay safe — a disciplined stack of factor regressions, rolling rank correlations, exceedance correlations, and a few hard exposure limits captures ~95% of the monitoring value at ~10% of the complexity. The real risk in a small multi-strategy book is not sophisticated dependence modelling; it is **hidden concentration** — several strategies silently loading on the same factor or ticker and then co-drawing down in a regime shift. Pod shops like Millennium and Citadel spend enormous resources on exactly this problem because they have already proven what the academic literature (Longin & Solnik 2001; Ang & Chen 2002; Embrechts, McNeil & Straumann 2002) predicts: **correlations during stress events are systematically higher than the full-sample Pearson estimate suggests**, especially in the left tail, and standard diversification math fails precisely when it is needed. What follows is a concrete, runnable framework — measurement stack, limit policy, factor decomposition, and alerting — calibrated for a single operator with Python, SQLite, and Alpaca.

Why Pearson alone will lie to you

The default instinct — compute a rolling Pearson correlation matrix of daily strategy P&Ls and watch it — will produce a number that is **both too optimistic in calm periods and too optimistic in crises**. Two well-documented problems undermine it. First, Pearson assumes linear dependence and finite second moments; equity strategy returns have fat tails (kurtosis typically 5-15 for short-horizon equity strategies), so a handful of crisis days dominates the estimate and makes the sample correlation unstable. Second, as Longin and Solnik (2001, *Journal of Finance*) showed using extreme value theory on the five largest equity markets from 1958-1996, **correlation rises sharply in bear-market tails but not in bull-market tails** ^{Longin} ^{ust} — they formally reject multivariate normality in the left tail. ^{SSRN} ^{ust} Ang and Chen (2002) documented the same asymmetry within the U.S. equity universe. ^{ResearchGate} ^{ScienceDirect} The practical consequence: a full-sample Pearson of 0.30 between two of your strategies can easily become 0.70+ on the ten worst days of the year, exactly when portfolio-level losses compound.

The cleanest fix at retail scale is a **three-layer measurement stack** rather than a single statistic. Use Spearman rank correlation as the primary day-to-day monitor

because it is invariant to monotone transformations, robust to outliers, [Wikipedia](#) and a consistent estimator of the underlying copula’s concordance (Embrechts, McNeil & Straumann 2002). Keep Pearson alongside it — when Pearson and Spearman diverge by more than ~0.15, the difference is a flag that a few extreme days are driving the dependence. Layer on an **exceedance correlation** — the Pearson correlation computed only on days where both strategies’ returns lie beyond a threshold, [ResearchGate](#) typically the 10th or 90th percentile of their marginal distribution. The Longin-Solnik formulation is:

$$\rho^+(\theta) = \text{Corr}(r_1, r_2 \mid r_1 > \theta, r_2 > \theta), \quad \rho^-(\theta) = \text{Corr}(r_1, r_2 \mid r_1 < -\theta, r_2 < -\theta)$$

With 250 trading days and a 10% threshold you get roughly 25 tail observations per strategy pair — Longin and Solnik themselves used 16-25 tail observations per country [Longin](#) — which is noisy but usable as a **comparative** signal: if the negative exceedance is materially higher than the positive exceedance, you have asymmetric tail dependence and should size more conservatively. For a 3-7 strategy book, compute it on a rolling 252-day window and refresh weekly.

DCC-GARCH (Engle 2002, JBES) is overkill for N=3-7 at retail scale. It requires fitting a univariate GARCH to each series plus two extra correlation-dynamics parameters, [Taylor & Francis Online](#) [SSRN](#) and its main payoff — time-varying conditional correlations — is already captured adequately by a 63-day rolling Spearman. The `arch` Python library (Kevin Sheppard) supports univariate GARCH cleanly [DataCamp](#) but full multivariate DCC typically requires R’s `rmgarch` or bespoke code; [GitHub](#) the engineering cost is not worth it for a solo operator. Use DCC only if you later scale to N > 10 or trade options where conditional covariance forecasts matter for hedging.

The copula / tail-dependence route (Embrechts-McNeil-Straumann 2002; Joe 1997) is elegant but practically unreliable at N=5 with 250 observations — the bivariate tail dependence coefficient $\lambda_U = \lim_{u \rightarrow 1^-} P(U_2 > u \mid U_1 > u)$ [bggj](#) has standard errors larger than its typical magnitude at small sample. **Skip formal copula fitting**; exceedance correlation gives you 80% of the tail-risk signal with a tenth of the estimation noise.

A minimum viable measurement stack

Metric	Window	Frequency	Purpose	Python tool
Spearman rank correlation	63 trading	Daily	Primary monitor; robust to	<code>scipy.stats.spearmanr</code>

matrix	days		outliers	
Pearson correlation matrix	63 trading days	Daily	Compare to Spearman; gap signals fat-tail drivers	<code>numpy.corrcoef</code>
Negative exceedance correlation	252 days, 10% threshold	Weekly	Detect asymmetric tail dependence	Custom (filter → Pearson)
Rolling factor betas	126 days (6-month), HAC SE	Weekly	Style drift, hidden common exposure	<code>statsmodels.OLS</code> with <code>cov_type='HAC'</code> Codes
Portfolio HHI / concentration	Real-time	Intraday	Ticker and sector concentration	Custom aggregation

Store results in SQLite using a schema like `correlation_matrices(date, method, strategy_a, strategy_b, value, window)` and `factor_loadings(date, strategy, factor, beta, tstat, r2)`. Indexing on `(date, strategy)` keeps daily updates under a second. This entire stack runs in under 50 lines of pandas/numpy and produces every signal a solo operator needs.

Factor decomposition is where hidden correlation hides

Two strategies can have Pearson correlation near zero on daily P&L and still be deadily correlated through a shared factor loading. A mean-reversion strategy on bank stocks and a trend strategy on regional REITs will look independent day-to-day but both short “quality” and long rate-sensitive value; a single hawkish Fed surprise hits both. Since Ross (1976) articulated APT, this has been the central reason institutional risk starts with factor exposure, not return correlation.

For 2-8 day hold strategies, use the **Carhart 4-factor model** (MKT, SMB, HML, UMD momentum; Carhart 1997 *Journal of Finance*) as the baseline rather than Fama-French 3 or 5, because short-horizon equity strategies frequently load on momentum. Augment with **QMJ** (Asness-Frazzini-Pedersen 2019, “Quality Minus Junk”) [Google Scholar](#) and optionally **BAB** (Frazzini-Pedersen 2014, “Betting Against Beta”) [ScienceDirect](#) if you

trade low-vol or quality sleeves. The Fama-French 5-factor model (RMW, CMA added) [QuantConnect](#) is better for long-horizon equity but its investment and profitability factors rebalance annually — they add noise at daily frequency with little benefit. Hou, Xue, and Zhang (2015) q-factor work [NBER](#) [Cheung Kong Graduate School of](#) reinforces that the 5-factor investment/profitability choices are somewhat arbitrary, so don't overfit to them.

Data sources are trivial: Ken French's library [Sec-api](#) at

mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html ships FF3, FF5, and UMD in daily, weekly, and monthly CSVs; AQR's datasets page

(aqr.com/Insights/Datasets) provides QMJ, BAB, and HML-Devil. [NYU Stern](#) The

[pandas-datareader](#) `famafrench` reader pulls them directly: [statsmodels](#)

[Pandas DataReader](#) `pandas_datareader.DataReader('F-F_Research_Data_5_Factors_2x3_daily', 'famafrench')` . For the regression, use OLS with **Newey-West HAC standard errors** to handle autocorrelation and heteroskedasticity: [Codes](#)

```
import pandas as pd, statsmodels.api as sm
from pandas_datareader import data as pdr

ff = pdr.DataReader('F-F_Research_Data_5_Factors_2x3_daily', 'famafrench')[0] / 1
mom = pdr.DataReader('F-F_Momentum_Factor_daily', 'famafrench')[0] / 100
factors = ff.join(mom).rename(columns={'Mkt-RF': 'MKT', 'Mom': 'UMD'})

def decompose(strategy_returns, factors, window=126):
    df = pd.concat([strategy_returns.rename('r'), factors], axis=1).dropna()
    df['xr'] = df['r'] - df['RF']
    X = sm.add_constant(df[['MKT', 'SMB', 'HML', 'UMD']])
    full = sm.OLS(df['xr'], X).fit(cov_type='HAC', cov_kws={'maxlags':5})
    rolling = pd.DataFrame(index=df.index, columns=['alpha', 'MKT', 'SMB', 'HML', 'UMD'])
    for i in range(window, len(df)):
        sub = df.iloc[i-window:i]
        m = sm.OLS(sub['xr'], sm.add_constant(sub[['MKT', 'SMB', 'HML', 'UMD']])).fit()
        rolling.iloc[i] = [m.params['const']] + list(m.params[1:])
    return full, rolling
```

The alpha test is simple: run the regression, look at the intercept's HAC t-statistic. If $|t| < 2.0$ the strategy isn't generating alpha — it's a factor exposure wearing a trading algorithm costume. R^2 below 0.10 means your strategy is mostly idiosyncratic (good — genuine alpha or noise). R^2 above 0.60 means you're mostly replicating a factor portfolio and paying Alpaca commissions to do so. Between those bounds, the **betas themselves** tell you where correlation lurks. If strategies A and B both load 0.4 on UMD with $|t| > 3$, they share a momentum exposure even if their daily

P&L correlation is 0.1. Detect style drift by flagging any rolling 126-day beta that moves more than 2 standard deviations from its trailing 252-day mean; pod shops call this “factor creep” and Balyasny’s post-2018 risk overhaul explicitly built volatility-targeted factor monitoring [Netinterest](#) as an alpha-generation tool, not just a defensive measure.

[Netinterest](#)

For retail, **sector/style ETF regression** is a practical tradeable proxy: regress strategy returns against SPY, QQQ, IWM (size), MTUM (momentum), [iShares](#) [BlackRock](#) [QUAL](#) (quality), USMV (low-vol), and the XLK/XLF/XLE/XLV/XLY/XLP/XLI/XLU/XLRE/XLB/XLC sector ETFs. These produce hedgeable betas — unlike Ken French factors, you can actually short MTUM to neutralize momentum exposure in your Alpaca account. Run this regression monthly as a secondary check on the Ken French decomposition.

Exposure limits — what pod shops do and what you can copy

Millennium, Citadel, Point72, and Balyasny [Wealthwise](#) have converged on remarkably consistent risk architectures, and the publicly reported numbers give a solid template even for a ~\$100K retail book. Millennium’s explicit, algorithmically-enforced rule set is: **5% drawdown from allocated capital triggers a 50% capital cut; 7.5% drawdown terminates the pod.** [Substack](#) [Substack](#) Point72 runs bespoke parameters per pod based on [Substack](#) the factor exposures native to each sector [Substack](#) (binary biotech event risk vs. rates convexity gets different treatment). [Substack](#) Citadel’s Portfolio Construction and Risk Group [Substack](#) reports directly to the CEO and runs continuous stress tests. [Substack](#) The universal features: hard drawdown stops, factor-neutrality requirements for the bulk of the book (with idiosyncratic sub-sector views permitted), real-time tracking of cross-pod factor overlap, [Substack](#) and daily capital reallocation away from underperformers. [Substack](#) [Substack](#)

Translated to a retail operator with 3-7 strategies, the minimum viable policy looks like this:

Hard limits (enforced by pre-trade checks, never overridden): single-name cap of 5-8% of total portfolio NAV across the book, regardless of which strategies hold it — the ticker is the unit of risk, not the strategy; GICS sector cap of 25-30% of NAV (matches typical institutional limits); maximum 1.5x gross leverage given Alpaca’s Reg T margin constraints; pattern day-trader rule (\$25K equity minimum for >3 day trades per 5 days) [FINRA +2](#) is a structural retail constraint to respect. Set a book-level drawdown circuit breaker at 8-10% from high-water mark — softer than Millennium’s 7.5% because you have concentration risk they don’t, so the buffer needs to be wider.

Soft limits (trigger review, not auto-liquidation): cross-strategy daily P&L

correlation (63-day Spearman) above 0.5 for any pair sustained 5+ trading days; factor beta on any single factor (MKT, SMB, HML, UMD, QMJ) above 0.5 aggregated across all strategies; 21-day rolling book volatility exceeding 150% of its 252-day average. These are warning flags that cause you to stop adding exposure in the correlated direction, not necessarily to cut.

The hard-vs-soft decision framework borrows directly from Millennium's architecture: **limits that protect against ruin are hard and enforced in code; limits that indicate edge degradation are soft and enforced by process.** A common retail mistake is to make correlation limits hard — this creates whipsaw because correlation estimates are noisy (a 63-day Spearman has a standard error of roughly $\frac{1}{\sqrt{63-3}} \approx 0.13\%$ under the null).

No-overlap vs soft-overlap on ticker: with 3-7 strategies and 5-10 total positions, **soft overlap with aggregate single-name cap is strictly better than no-overlap.** Hard no-overlap wastes alpha when two strategies independently generate the same signal (which is evidence, not coincidence), and the one-ticker-per-strategy rule forces artificial substitution into worse expressions of the same idea. The aggregate cap handles the concentration risk directly. The exception: wash-sale rules make hard no-overlap attractive for any ticker you're actively tax-loss harvesting in one strategy while another strategy wants to re-buy within 30 days (Fidelity) — the IRS doesn't care about your strategy labels. (Vanguard)

The **Statman (1987)** and **Evans-Archer (1968)** findings that you need 30-40 stocks for idiosyncratic diversification (Efficientfrontier +5) apply to *passive* portfolios; they don't translate directly to active multi-strategy books. More relevant is **Ledoit & Wolf (2003, 2004)** on shrinkage estimation: with only 5-10 positions and 252 daily observations, your sample covariance matrix is ill-conditioned and mean-variance optimization will produce extreme, unstable weights. (SSRN) (IDEAS/RePEc) The practical fix is **equal-risk-contribution sizing** (risk parity within each strategy) combined with shrinkage — `sklearn.covariance.LedoitWolf` does the shrinkage automatically and is a one-line substitute for `np.cov`. Don't try to run Markowitz optimization on 5-10 positions; the sampling error on expected returns exceeds the expected returns themselves.

Alerting and change-point detection without alert fatigue

The single biggest operational failure mode for a solo operator is **alert fatigue**: you set up Telegram notifications, get 40 alerts in the first week, start muting them, and then miss the real one. The research on alerting (transferable from medical telemetry and aviation cockpits) consistently shows that alerts must be tiered (Security 101) (PagerDuty) and the CRITICAL tier reserved for events that demand action within minutes. A sensible three-tier structure:

- **INFO** (daily digest email at market close): factor loadings, correlation heatmap, P&L attribution, current HHI. No Telegram push.
- **WARN** (Telegram, business hours only): rolling 30-day correlation between any pair > 0.5 for 5 consecutive days; factor beta change $> 2\sigma$ from trailing mean; sector concentration $> 25\%$; single-name position $> 6\%$.
- **CRITICAL** (Telegram with retry, 24/7): hard limit breach (single-name $> 8\%$, book drawdown $> 8\%$); any strategy down $> 3\%$ intraday; connectivity/execution failure.

Use `python-telegram-bot` for messaging with deduplication (hash the alert content, suppress if same hash fired in last 60 minutes) and a snooze command so you can silence a known WARN during a planned rebalance. For the daily digest, Streamlit beats Dash for a solo operator — simpler, [Substack](#) single-file deployment, [Kanaries](#) good for private use on Tailscale or a Fly.io instance.

For change-point detection on correlation and factor-loading time series, the `ruptures` library (Truong, Oudre & Vayatis 2020, *Signal Processing*) is the clear choice. It implements PELT (Killick, Fearnhead, Eckley 2012), binary segmentation, window-based methods, and kernel-based detection in a consistent API. [PyPI](#)

[Centre-borelli](#) For a retail monitor, run

```
rpt.Pelt(model="rbf").fit(signal).predict(pen=penalty)
```

[Centre-borelli](#)

[GitHub](#)

weekly on the rolling beta time series for each strategy/factor pair. The RBF kernel detects distribution changes, not just mean shifts, which is exactly what regime changes in factor exposure look like. Calibrate the penalty empirically on your historical data — too low and it finds phantom breakpoints, too high and it misses real ones. Typical starting value: $\text{pen} = 3 * \sigma^2 * \log(n)$ where σ is the series standard deviation.

Bayesian Online Change Point Detection (Adams & MacKay 2007, arXiv:0710.3742)

[ACL Anthology](#) is the streaming alternative [DeepWiki](#) — it produces a posterior

probability [Princeton](#) of a change point having occurred at each step, [arXiv](#) [Auai](#)

which is more natural for real-time alerting than PELT's offline segmentation. The

`bayesian-change-point-detection` package implements it. [GitHub](#) [CRAN](#) For a solo

operator running weekly batch jobs, `ruptures` + PELT is simpler and adequate; reserve BOCPD for if you later move to intraday monitoring.

Specific numeric triggers that have survived retail practitioner scrutiny: cross-strategy 30-day rolling Spearman crossing 0.5 and staying there for 5 days (CUSUM-style persistence check reduces false positives); any factor beta moving > 0.25 in absolute terms over 20 trading days; 21-day realized volatility exceeding $1.5\times$ the 252-day average; portfolio VaR (historical simulation at 95%) increasing $> 50\%$ week-over-week. Each of these maps to a known failure mode — correlation regime shift, style drift, vol

regime change, and tail thickening respectively.

A retail operator's decision rubric

The hardest judgment calls in this architecture are not statistical but **organizational**. Three rules keep a solo operator honest.

First, **factor decomposition runs weekly, not daily** — the signal-to-noise ratio on daily factor betas is terrible and you will whipsaw yourself. Weekly rebalancing of monitoring (not of positions) matches the statistical reliability of the estimates.

Second, **every strategy must publish its expected factor profile before it goes live**, and the monitoring system compares realized factor loadings to the expected profile. A momentum strategy should load positive on UMD; if its realized UMD beta is zero, something is broken in the implementation. This catches bugs that P&L alone will miss until it's too late.

Third, **never override a hard limit during a drawdown**. The pod-shop literature is unanimous on this: Millennium's 15-20% annual PM turnover [Netinterest](#) is not a bug, it's the feature [Substack](#) [Substack](#) that prevents catastrophic losses. [Substack](#) At retail, the equivalent discipline is to pre-commit to cutting a strategy's allocation by 50% on a 5% strategy-level drawdown and shutting it off at 8%, regardless of your conviction about the setup. The behavioral literature (Shefrin & Statman's mental accounting work) predicts exactly the bias you'll face — wanting to "give it one more week" — and the rule exists specifically to override that bias.

Conclusion

The monitoring stack that actually works for a 3-7 strategy retail book on Alpaca is unglamorous: Spearman correlation on rolling windows, Carhart+QMJ factor decomposition with HAC standard errors, exceedance correlation on stressed days, PELT change-point detection on betas, hard caps at the ticker and sector level, and tiered Telegram alerts. Every piece of it runs in standard Python libraries (`scipy` , `statsmodels` , `pandas-datareader` , `ruptures` , `python-telegram-bot`) and stores cleanly in SQLite. The sophisticated techniques — DCC-GARCH, copula fitting, full BARRA-style risk models — offer diminishing returns at this scale and introduce estimation noise that outweighs their theoretical advantages.

The insight worth internalizing is that the pod shops' edge is not their statistics; it's their enforcement. Millennium's 5%/7.5% drawdown rules are not better than a thoughtful retail operator's 5%/8% rules — they're just executed without exception. A solo trader who builds this stack and actually obeys the limits will outperform a solo

trader with a more sophisticated model who overrides limits under stress. Build the measurement system once, write the policy document, automate the enforcement, and then spend your analytical time finding new alpha rather than refining the risk architecture. The empirical literature has been telling us for 25 years that tail correlations explode in crises — the only question is whether you have a limit structure that fires before your P&L does.